SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	04-08-2026	**Duration:**	60 minutes

Code of Conduct

*I **K Madan Mohan Reddy (192411206**) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _________**Byreddy**________________

Annexure A – Architecture Design Assignment

Instructions to Students:

"Online Loan Eligibility Prediction and Approval Management Platform"

1. Analyze System Requirements

- **Study the given problem statement and identify the system objectives.**

  The Online Loan Eligibility Prediction and Approval Management Platform is designed to automate and simplify the loan application and approval process for both customers and financial institutions. Its primary objective is to predict an applicant's loan eligibility based on factors such as income, employment status, credit history, loan amount, and repayment capacity. The system enables users to apply for loans online, upload the required documents, and track the status of their applications in real time. It also helps loan officers process applications more efficiently by reducing manual work, minimizing errors, and supporting faster approval decisions. Additionally, the platform securely stores customer and loan data, ensures data privacy, generates reports for administrative analysis, and provides a user-friendly experience that improves the overall efficiency, accuracy, and transparency of the loan management process.

- **Analyze the functional and non-functional requirements that influence the software architecture.**
- The Online Loan Eligibility Prediction and Approval Management Platform is developed to automate and streamline the loan application and approval process for both customers and financial institutions. The primary objective of the system is to accurately predict an applicant's loan eligibility by analyzing factors such as income, employment status, credit history, loan amount, and repayment capacity.

- It allows customers to submit loan applications online, upload the necessary documents, and monitor the status of their applications in real time. The platform also assists loan officers by automating eligibility verification, reducing manual effort, minimizing processing errors, and accelerating approval decisions.

- Furthermore, the system securely manages customer and loan data, ensures data privacy and confidentiality, generates analytical reports for administrators, and provides a user-friendly interface that enhances the overall efficiency, accuracy, transparency, and reliability of the loan management process.

- **Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security**.

  The Online Loan Eligibility Prediction and Approval Management Platform should satisfy several key quality attributes to ensure it is efficient, reliable, and secure. Scalability enables the platform to handle a growing number of users, loan applications, and transactions without affecting performance. Maintainability ensures that the system can be easily updated, modified, and enhanced to meet changing business requirements and regulations. Reliability guarantees that the platform consistently produces accurate loan eligibility predictions and processes applications with minimal errors.

  - Availability ensures that the system remains accessible to customers and administrators whenever needed, minimizing downtime through robust infrastructure.
  - Performance focuses on providing fast response times for application submission, eligibility prediction, document verification, and approval processing, even during periods of high demand.
  - Security protects sensitive customer information and financial data through secure authentication, authorization, data encryption, and regular security measures, ensuring confidentiality, integrity, and compliance with data protection standards.
  - These quality attributes collectively contribute to a dependable, efficient, and user-friendly loan management platform.

2. Select a Suitable Software Architecture

- **Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).**

  A Layered Architecture combined with the Model–View–Controller (MVC**)** architectural pattern is an appropriate choice for the Online Loan Eligibility Prediction and Approval Management Platform. The layered architecture separates the system into distinct layers, such as the presentation layer, business logic layer, data access layer, and database layer, making the application easier to develop, maintain, test, and scale.

  - The MVC pattern further organizes the application by separating the user interface (View), application logic (Controller), and data management (Model), improving code reusability and maintainability.
  - Users interact with the presentation layer to submit loan applications and view their status, while the business logic layer validates applicant information, predicts loan eligibility, and manages the approval workflow.

- The data access layer handles communication with the database, where customer details, loan applications, and approval records are securely stored.
- This architectural approach enhances modularity, security, performance, and maintainability, making it well suited for a web-based loan management system.

- **Justify why the selected architecture is the most suitable for the given problem.**

  The Three-Tier Architecture is the most suitable architecture for an Online Loan Eligibility Prediction and Approval Management Platform because it provides a clear separation between the user interface, business logic, and database, making the system more organized, secure, and easier to maintain. The presentation layer allows customers, loan officers, and administrators to interact with the platform through user-friendly interfaces, while the business logic layer processes loan applications, verifies applicant details, applies eligibility rules, and integrates the machine learning model to predict loan eligibility. The data layer securely stores customer information, loan records, repayment history, and prediction results. This architecture enhances security by preventing direct access to the database and supporting authentication and authorization mechanisms. It also improves scalability, allowing the application to handle a large number of users and transactions efficiently.

  ✔ Furthermore, updates to loan policies or eligibility criteria can be made in the business layer without affecting the user interface or database, reducing maintenance effort.

  ✔ The modular design also supports future enhancements such as fraud detection, mobile applications, payment integration, and notification services.

  ✔ Overall, the three-tier architecture offers high performance, flexibility, reliability, and security, making it the ideal choice for developing a modern online loan eligibility prediction and approval management platform.

3. Draw the Software Architecture Diagram

- **Design a clear architecture diagram illustrating the overall structure of the proposed system**.

```
            ONLINE LOAN ELIGIBILITY PREDICTION

            AND APPROVAL MANAGEMENT PLATFORM



+------------------------------------------------------------+

|                  PRESENTATION LAYER                  |

|------------------------------------------------------------|

|  Customer Portal | Loan Officer Portal | Admin Dashboard     |
```

```
|  - Register/Login                          |
|  - Apply for Loan                          |
|  - Upload Documents                        |
|  - Track Application Status                |
+--------------------------▲---------------------------------+
                    |
                    | HTTP/HTTPS
                    ▼
+-----------------------------------------------------------+
|               BUSINESS LOGIC LAYER                  |
|-----------------------------------------------------------|
| • User Authentication & Authorization               |
| • Loan Application Management                       |
| • Document Verification                             |
| • Loan Eligibility Prediction (ML Model)            |
| • Credit Score Analysis                             |
| • Loan Approval & Rejection Workflow                |
| • Notification Service (Email/SMS)                  |
| • Report Generation                                 |
+--------------------------▲---------------------------------+
                    |
                    | SQL/API
                    ▼
+-----------------------------------------------------------+
|                   DATA LAYER                        |
```

```
|------------------------------------------------------------|
| • Customer Database                                 |
| • Loan Application Database                         |
| • Credit History Records                            |
| • ML Prediction Data                                |
| • Document Repository                               |
| • Audit Logs                                        |
+------------------------------------------------------------+
```

External Services

• Credit Bureau API

• Email/SMS Gateway

• Payment Gateway

Include all major layers/components, databases, external systems, APIs, and communication mechanisms.

- **Use appropriate software architecture notations and labels.**

```
+--------------------------------------+
|            <<Actor>>             |
|  Customer / Loan Officer / Admin   |
+------------------+-------------------+
                   |
            HTTPS/REST API
                   |
                   v
```

```
+------------------------------------------------------------------+
|                <<Presentation Layer>>                      |
|------------------------------------------------------------------|
| • Web Application (UI)                                    |
| • Mobile Application                                      |
| • Login & Registration                                    |
| • Loan Application Form                                   |
| • Document Upload                                         |
| • Application Status Dashboard                            |
+------------------------------+-----------------------------------+
                               |
                        Service Requests
                               |
                               v
+------------------------------------------------------------------+
|                <<Business Logic Layer>>                    |
|------------------------------------------------------------------|
| <<Authentication Module>>                                 |
| <<Loan Management Module>>                                |
| <<Eligibility Prediction Module (ML)>>                    |
| <<Document Verification Module>>                          |
| <<Approval Management Module>>                            |
| <<Notification Module>>                                   |
| <<Reporting Module>>                                      |
+------------------------------+-----------------------------------+
```

4. Explain Components and Their Interactions

- **Describe the purpose and responsibilities of each architectural component**.

  The architecture of the Online Loan Eligibility Prediction and Approval Management Platform consists of several components, each with a specific purpose and responsibility. The Presentation Layer serves as the user interface, allowing customers, loan officers, and administrators to register, log in, submit loan applications, upload documents, and track application status. It also validates user inputs before forwarding requests to the system. The Business Logic Layer is the core of the application, responsible for authenticating users, processing loan applications, verifying documents, executing the machine learning model for loan eligibility prediction, applying loan approval rules, generating reports, and sending notifications. The Data Layer securely stores and manages customer information, loan applications, credit history, uploaded documents, prediction results, and audit logs while ensuring data integrity and efficient retrieval.

- **Explain how the components communicate and exchange data.**

  The components of the Online Loan Eligibility Prediction and Approval Management Platform communicate and exchange data through well-defined interfaces and service requests. The Presentation Layer communicates with the Business Logic Layer using secure HTTP/HTTPS requests or REST APIs. When a user submits a loan application, the presentation layer collects the applicant details and sends the data to the business logic layer for processing. The Business Logic Layer validates the information, applies loan approval rules, communicates with the Machine Learning Prediction Module to evaluate eligibility, and generates the approval or rejection decision. The prediction module receives applicant data such as income, credit score, employment details, and loan requirements, then returns an eligibility score or prediction result to the business layer. The business layer communicates with the Data Layer using database queries or data access services to store and retrieve customer profiles, loan applications, credit records, documents, and transaction details

- **Illustrate the overall workflow of the system.**

  The workflow of the system begins when a customer accesses the platform and creates an account or logs in. The customer then submits a loan application by providing personal details, employment information, income details, loan requirements, and required documents. The submitted information is transferred from the Presentation Layer to the Business Logic Layer for processing. The business layer authenticates the user, validates the application details, and verifies the uploaded documents. The system then communicates with external services such as credit bureaus to obtain credit scores and financial history. The collected applicant data is passed to the Machine Learning Prediction Module, which analyzes the information and generates a loan eligibility prediction.

✔ Based on the prediction result and predefined approval rules, the system decides whether the loan application should be approved, rejected, or reviewed manually. The decision and related information are stored securely in the Data Layer.

- ✔ Finally, the customer and loan officers are notified about the application status through email or SMS services. Administrators can monitor applications, manage users, and generate reports through the administrative dashboard.

5. Discuss Quality Attributes
Evaluate how the proposed architecture addresses the following aspects:

- **Scalability**
- The architecture supports scalability by separating the presentation, business logic, and data layers. Each layer can be expanded independently based on system requirements. For example, additional application servers can be added to the business logic layer to handle increased loan applications and user requests without affecting other components. This allows the platform to support a growing number of customers and transactions.

- **Security**

  The architecture improves security by restricting direct access to sensitive data through the separation of layers. User authentication and authorization are handled in the business logic layer, while databases are protected from direct user interaction. The system can implement encryption, secure API communication, role-based access control, and audit logging to protect confidential financial information.

- **Performance**

  The architecture enhances performance by dividing responsibilities among different components. The business logic layer processes loan eligibility calculations and machine learning predictions efficiently, while the data layer is optimized for storing and retrieving information. Techniques such as database indexing, caching, and optimized API communication can further improve response time and overall system efficiency.

- **Reliability**

  The modular structure of the architecture increases reliability because failures in one component do not necessarily affect the entire system. For example, issues in the user interface layer can be resolved without impacting the database or prediction module. Error handling, backup mechanisms, and transaction management ensure consistent and reliable loan processing.

- **Maintainability**

  The separation of concerns makes the system easier to maintain and update. Changes to loan approval rules, machine learning models, or user interfaces can be performed independently without modifying the complete system. This reduces development effort and allows faster implementation of new features and policy changes.

- **Availability**

  The architecture supports high availability by allowing components to be deployed separately and supported with backup and recovery mechanisms. Load balancing, database replication, and fault-tolerant services can be implemented to minimize downtime and ensure that users can access loan services continuously.

6. Justify Architectural Decisions

- **Explain the rationale behind your architectural choices.**

  The Three-Tier Architecture was selected for the Online Loan Eligibility Prediction and Approval Management Platform because it provides a clear separation of responsibilities between the user interface, business processing, and data management layers. This separation is suitable for a financial application where security, accuracy, scalability, and maintainability are critical. The architecture allows the presentation layer to focus on user interaction, the business layer to handle loan processing and machine learning-based eligibility prediction, and the data layer to securely manage customer and loan-related information. This structure improves system organization and makes it easier to develop, test, and manage the application.

- **Discuss the trade-offs considered while selecting the architecture.**

  While selecting the architecture, several trade-offs were considered. The three-tier approach improves flexibility, security, and maintainability, but it may introduce additional complexity compared to a simple single-layer application because communication between layers requires additional processing. The separation of components may also increase initial development time and require proper design of interfaces between layers. However, these disadvantages are outweighed by the benefits of better scalability, easier updates, and improved control over sensitive financial data.

- **Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.**

  The chosen architecture supports future enhancements by allowing new features to be added without major changes to the entire system. For example, advanced artificial intelligence models, automated fraud detection, chatbot support, mobile applications, and improved reporting features can be integrated by extending the business logic layer. The system can also integrate with external services such as credit bureaus, payment gateways, identity verification systems, and banking platforms through APIs without affecting the core application structure.

    - For long-term maintainability, the modular design allows individual components to be updated, replaced, or upgraded independently.
    - Changes in loan policies, eligibility criteria, database technologies, or machine learning models can be implemented with minimal impact on other components.
    - This makes the platform adaptable to changing business requirements and technological advancements.
    - Overall, the selected architecture provides a balanced solution that supports current system requirements while ensuring future growth, integration capabilities, and long-term sustainability.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all major components, interfaces, and interactions accurately represented.	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and lacks sufficient detail.	Diagram is incomplete, unclear, or technically incorrect.
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical	Provides strong justification for architectural	Provides adequate justification with	Limited justification with minimal	Justification is weak or absent; report lacks

Presentation (10%)	decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.	minor gaps; report is well organized.	discussion of trade-offs or future scope; report needs improvement.	organization and technical clarity.

Criteria	**Mark**
System Requirement Analysis **(30%)**	/5
Selection of Software Architecture **(25%)**	/5
Architecture Diagram and Component Design **(20%)**	/5
Component Interaction and Quality Attribute Analysis **(15%)**	/5
Architectural Justification and Technical Presentation **(10%)**	/5
TOTAL	/25